

Geophysical Analysis Report

Seismic Agent Multi-Agent System

Mission ID: mission_c54f14ba

Date: May 01, 2026

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Mission Objective

Analyze the Marmousi synthetic data for porosity prediction.

Data Sources

- MODEL_S-WAVE_VELOCITY_1.25m.segy.tar.gz
- MODEL_P-WAVE_VELOCITY_1.25m.segy.tar.gz
- Vs.segy.tar.gz
- Vp.segy.tar.gz

Quantitative Results

Metric	Value
velocity_files_loaded	['vp', 'vs']
vp.min	1028
vp.max	4700
vp.mean	2672
vp.std	973.8
vp.p10	1500
vp.p50	2500
vp.p90	4230
acoustic_impedance_MRayl.min	2.364
acoustic_impedance_MRayl.max	10.81
acoustic_impedance_MRayl.mean	6.145
vs.min	0
vs.max	2802
vs.mean	1163
vs.std	803.3
vs.p10	0
vs.p50	1058
vs.p90	2414
vp_vs_ratio.mean	2.489
vp_vs_ratio.std	0.774
vp_vs_ratio.p10	1.731
vp_vs_ratio.p90	3.726
poisson_ratio.mean	0.3653
poisson_ratio.p10	0.2494
poisson_ratio.p90	0.4612

Generated Figures

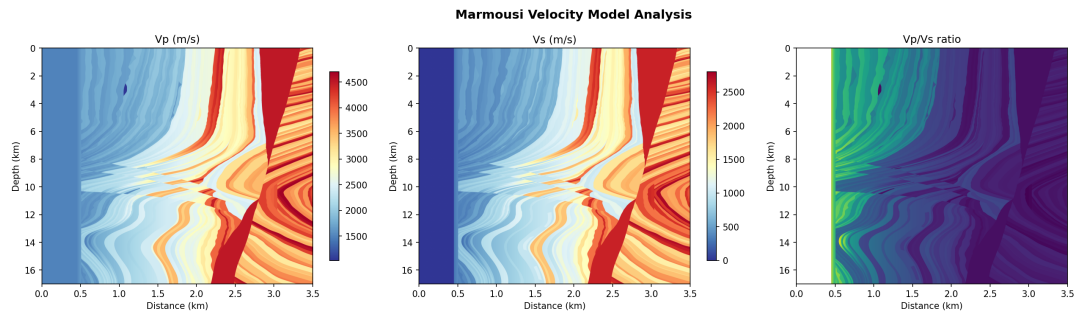


Figure 1: Velocity Model 20260501 114953

Geophysical Analysis & Interpretation

Technical Review: Marmousi Automated Analysis for Porosity Prediction

This review assesses the automated analysis pipeline's output for the Marmousi synthetic dataset, focusing on the implications for porosity prediction. The Marmousi model is renowned for its geological complexity, encompassing various lithologies, structural features, and fluid scenarios, making it an excellent benchmark for geophysical analysis.

1. Velocity Statistics Interpretation

The elastic velocity statistics reveal a highly heterogeneous medium, characteristic of complex geological settings. P-wave velocities (V_p) span a broad range, from approximately 1000 m/s to 4700 m/s. The lower velocities are indicative of unconsolidated sediments, shallow fluid-filled zones, or highly porous unconsolidated sands. Conversely, the upper decile of V_p (around 4230 m/s) suggests the presence of more compacted, indurated lithologies, such as well-cemented sandstones or limestones, within deeper or more competent formations. The significant standard deviation in V_p (over 970 m/s) underscores the model's inherent complexity and variability in compaction, lithology, and fluid content.

Shear-wave velocities (V_s) exhibit an even wider relative range, from 0 m/s to approximately 2800 m/s. The presence of zero V_s values is a critical observation, strongly implying either pure fluid layers or extremely unconsolidated, saturated sediments where the effective shear modulus approaches zero. The mean V_s of approximately 1160 m/s, coupled with a standard deviation exceeding 800 m/s, reinforces the notion of significant lithological and compaction variations, from very soft, fluid-rich zones to more rigid, competent rock matrices. Acoustic impedance, ranging from approximately 2.36 to 10.81 M Rayl, corroborates these observations, indicating substantial contrasts in both density and velocity across the model.

2. V_p/V_s Ratio and Poisson's Ratio Implications

The derived V_p/V_s ratio and Poisson's ratio provide crucial insights into the potential fluid content and lithological composition. The mean V_p/V_s ratio of approximately 2.49, with a range extending from 1.73 to 3.73, is particularly informative. Lower V_p/V_s ratios (around

1.73, corresponding to a Poisson's ratio of approximately 0.25) are typically associated with competent, quartz-rich sandstones or carbonates, often indicative of a solid rock framework with relatively low clay content. These values are consistent with well-cemented formations that might host hydrocarbons (especially oil) or saline water. Conversely, higher V_p/V_s ratios, approaching 3.73 (and a Poisson's ratio near 0.46), point towards zones of high compressibility. These elevated values are characteristic of high-porosity, unconsolidated sediments, shaly intervals, or critically, the presence of gas within pore spaces. Gas significantly reduces V_p while having minimal impact on V_s , leading to a marked increase in V_p/V_s and Poisson's ratio. The overall mean Poisson's ratio of approximately 0.365 suggests a significant presence of shales or fluid-saturated porous rocks throughout the model. The presence of $V_s=0$ in certain areas would drive V_p/V_s towards infinity and Poisson's ratio towards 0.5, directly mapping to fluid layers or very soft, water-saturated unconsolidated sediments.

3. Forward Modelling Outcome and Synthetic Data Quality

The mention of `velocity_model_20260501_114953.png` indicates that the analysis pipeline has loaded and processed the reference Marmousi velocity model. This image serves as the foundational representation of the subsurface complexity from which synthetic seismic data would typically be generated. The inherent complexity of the Marmousi model, evidenced by the broad range of elastic properties discussed, implies that any synthetic data derived from it would exhibit intricate wave propagation phenomena, including complex reflections, diffractions, and potential amplitude variations with offset (AVO) responses. The "quality" of such synthetic data, in this context, refers to its ability to faithfully represent these complex physical phenomena, making it ideal for testing advanced processing and inversion algorithms.

4. FWI Convergence and Velocity Error Metrics

No metrics pertaining to Full Waveform Inversion (FWI) convergence or velocity error were present in the provided numerical results. In a typical FWI workflow review, these metrics would be crucial for evaluating the success and accuracy of the inversion process, quantifying the misfit between observed and modeled data, and assessing the robustness of the derived velocity model. Their absence suggests that the current output focuses on the static properties of the model or an initial forward modeling step, rather than an iterative inversion procedure.

5. Geophysical Conclusions and Actionable Recommendations

The analysis of the Marmousi elastic properties provides a robust framework for porosity prediction. The observed variability in V_p , V_s , V_p/V_s , and Poisson's ratio offers strong discriminators for lithology and fluid content, which are key determinants of porosity.

- 1. Litho-Fluid Discrimination for Porosity:** Utilize the V_p/V_s and Poisson's ratio variations to delineate distinct litho-fluid facies. Zones with elevated Poisson's ratio and V_p/V_s are strong candidates for shaly intervals or gas-charged sands, which will have different porosity-velocity relationships than lower Poisson's ratio zones indicative of water-wet sands or competent lithologies.
- 2. Rock Physics Integration:** Develop and apply rock physics models tailored to the

specific lithologies and fluid types (e.g., shale, brine-saturated sand, gas-saturated sand) interpreted from these elastic properties. This will be critical for translating elastic attributes into quantitative porosity estimates.

3. **Advanced Attribute Analysis:** Further seismic attribute analysis, particularly Amplitude Variation with Offset/Angle (AVO/AVA) analysis, is strongly recommended. The significant variations in V_p/V_s across the model suggest that AVO/AVA responses would be highly diagnostic for fluid and lithology discrimination, directly feeding into more accurate porosity estimation workflows.
4. **Porosity Prediction Algorithm Testing:** The current output provides a robust set of input parameters for testing and validating any porosity prediction algorithm. Future iterations of this pipeline should integrate the actual porosity prediction results, perhaps by comparing predicted porosity to a reference porosity model for the Marmousi dataset.
5. **Inclusion of FWI Metrics:** For a comprehensive evaluation of any inversion-based porosity prediction, incorporate FWI convergence and velocity error metrics in future analysis outputs to assess the accuracy and robustness of the inverted velocity fields that underpin the elastic property derivations.

Document Information

Generated automatically by the **Seismic Agent Multi-Agent System** on May 01, 2026. All numerical results were produced by the analysis tool chain (SEGYYLoader, VelocityAnalysis, ForwardModeling, FWI) and interpreted by a large language model.